

3.9 DIFFUSION-WEIGHTED IMAGING (DWI) & ADC

SEEING THE MICROWORLD: WATER MOTION REVEALS PATHOLOGY

DWI detects the random (Brownian) motion of water molecules in tissue. Restricted diffusion often indicates cellularity, ischemia, or dense structures. ADC quantifies diffusion.

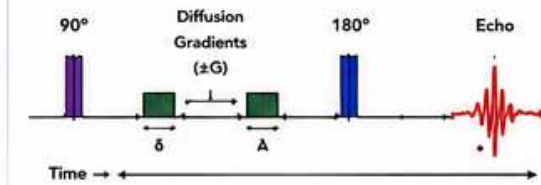


KEY CONCEPT

- DWI = sensitive to diffusion (restricted = bright).
- ADC = quantitative map (restricted = dark).
- High b-value → more sensitive to restriction.
- Pair DWI with ADC to avoid "T2 shine-through".

- ↑ Increase / Benefit
↓ Decrease / Trade-off
— No / Minimal Effect

1 HOW DWI WORKS (BASICS)



- Two equal and opposite diffusion gradients around the 180° pulse.
- Stationary spins refocus; moving spins don't → signal loss.
- The more diffusion, the more signal loss.
- b-value determines diffusion weighting.

★ DWI is exquisitely sensitive to diffusion but also to artifacts.

2 KEY PARAMETERS & WHAT THEY DO

Parameter	What It Does	Increase (↑)	Decrease (↓)
b-value (s/mm ²)	Strength of diffusion weighting	↑ Sensitivity to restriction	↓ Sensitivity, more T2 effect
TE	Longer TE increases diffusion weighting	↑ Diffusion contrast	↓ SNR
TR	Overall scan time & recovery	↑ SNR (longer)	↓ Scan time (shorter)
Number of Averages (NEX)	Improves SNR	↑ SNR	↓ SNR
Slice Thickness / Voxel Size	Affects SNR & partial volume	↑ SNR (thicker)	↓ Resolution
Fat Suppression	Reduces fat signal for cleaner lesions	↑ Lesion conspicuity	—

Common b-values: 0, 50–100, 500, 800–1000, 1500 s/mm²

3 IMAGE APPEARANCE (BRAIN EXAMPLES 1.5T)

T2-weighted (FSE)	DWI (b = 1000)	ADC Map
• CSF: Bright • Edema: Bright • White Matter: Intermediate • Cortical Gray: Intermediate • Lesion (subacute ischemia): Variable	• Normal brain: Dark • Acute infarct: Bright (restricted diffusion) • Abscess: Bright • Tumor (high cellular): Bright	• Normal brain: Gray • Acute infarct: Dark (low ADC) • Abscess/Tumor: Dark • T2 shine-through: Bright on DWI, bright on ADC

Always interpret DWI with ADC. Restricted = Bright DWI + Dark ADC.

5 DWI vs. ADC: HOW TO INTERPRET

Finding	DWI (High b)	ADC Map	Interpretation
Restricted diffusion	Bright	Dark	True restriction (cellular, ischemia, pus)
T2 shine-through	Bright	Bright/Gray	Not restricted (e.g., edema, tumor with high T2)
Facilitated diffusion	Dark	Bright	Increased diffusion (e.g., CSF, chronic infarct)
Normal tissue	Dark	Gray	No restriction

Use both maps together to avoid false positives.

6 COMMON ARTIFACTS & PITFALLS

Artifact / Pitfall	Cause	How to Reduce / Fix
T2 Shine-Through	Long T2 tissues appear bright on DWI	Always check ADC; if ADC bright, it's not restriction.
Eddy Currents (Geometric distortion)	High b-values, long echo train	Use parallel imaging, shorter TE, reduce echo spacing.
Susceptibility (geometric distortion)	Air-tissue interfaces, metal, sinus	Use higher bandwidth, parallel imaging, shorter echo train.
Motion	Patient motion, pulsation, breathing	Use faster sequences, immobilize, respiratory gating if needed.
Chemical Shift	Fat-water misregistration	Use fat suppression, high bandwidth.
Low SNR	High b-value, small voxel	Increase NEX, thicker slices, optimize coil.

7 b-VALUE GUIDE (GENERAL)

b-value (s/mm ²)	Use / Purpose	Pros	Cons
0	Reference (no diffusion weighting)	High SNR	No diffusion info
50–100	Low b (perfusion sensitive)	Good SNR, perfusion help	Less specific restriction
500	Standard clinical DWI	Good balance	—
800–1000	High b (restriction sensitive)	Better lesion detection	Lower SNR
1500–2000	Very high b	Very sensitive to restriction	Low SNR, more artifacts

For brain: b = 0 and 1000 are most common. For body: 0, 50, 400–800.

8 ENGINEER CHECKLIST (DWI)

- ✓ Is fat suppression working well?
- ✓ Are b-values correct and in the right order?
- ✓ Are diffusion gradients properly calibrated?
- ✓ Is distortion acceptable? Check EPI bandwidth.
- ✓ Is coil element functioning (no dropouts)?
- ✓ Is patient motion minimized?
- ✓ Do DWI and ADC align (no misregistration)?
- ✓ Review both DWI and ADC before concluding.

QUICK REFERENCE

Restricted Diffusion =
Bright DWI
+ Dark ADC

T2 Shine-Through =
Bright DWI
+ Bright ADC

9 CLINICAL EXAMPLES (BRAIN)

	Acute Ischemic Stroke	Bright DWI, Dark ADC (within minutes).
	Brain Abscess	Bright DWI, Dark ADC (pus restricts).
	Tumor (High Grade)	Often bright DWI, low ADC (high cellularity).
	Posterior Reversible Encephalopathy (PRES)	Usually no restriction (DWI iso/dark, ADC normal/bright).
	Chronic Infarct	Dark DWI, Bright ADC (facilitated).

10 CLINICAL EXAMPLES (BODY)

	Liver Metastasis	Often restricted (bright DWI, low ADC).
	Pancreatic Cancer	Restricted diffusion common.
	Prostate Cancer	Restricted diffusion in dominant lesions.
	Bone Marrow Lesions	Helps detect and characterize infiltration.
	Inflammation / Abscess	Restricted diffusion in pus collections.

11 PARAMETERS THAT MATTER MOST

Parameter	Impact on DWI	Recommendation
b-value	Most important	Use high b (800–1000) for brain.
Fat Suppression	Improves lesion conspicuity	Use STIR or SPIR/Dixon.
TE	Too long → low SNR	Keep as short as possible.
Bandwidth	Higher = less distortion, lower SNR	Use high BW for diffusion.
Parallel Imaging	Reduces distortion	Use 2–3x if available.
NEX	Improves SNR	Increase if time allows.

12 PRACTICAL TIPS

- Start with b = 0 and 1000 for brain.
- Use 50–800 for body depending on application.
- Check ADC early: it may reveal artifacts or issues.
- Short TE, high bandwidth, and parallel imaging reduce distortion.
- Maintain patient comfort—motion ruins DWI.
- Compare with T2/FLAIR and post-contrast images.



REMEMBER

DWI is about water motion.

- Restricted = cellular, ischemic, or pus.
- ADC is your truth detector.
- Bright DWI + Dark ADC = true restriction.



KEY TAKEAWAY

DWI/ADC is one of the most powerful tools in MRI.

Use the right b-values, control artifacts, and always interpret both DWI and ADC together.